Does a paraglider move differently in a group?

Group time against solo flight, on open flights

November 2026

Three questions

1. Do flights in a group move differently from solo flights?
2. Is it the group, or the route, season, hour and wing that go with it?
3. Does the same flight change when it enters a group, and when it leaves?

One thread: how straight a flight is, and how it climbs and glides.

FFVL archive, paragliders.

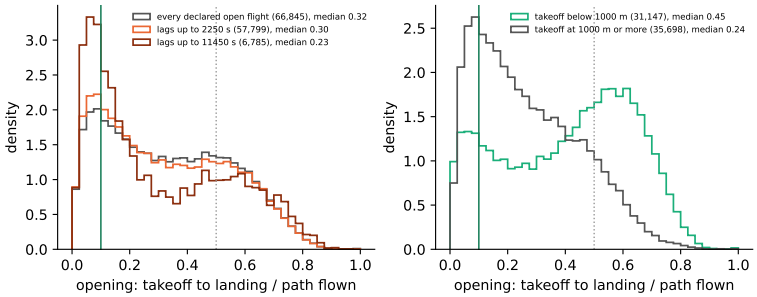
Three populations of flight time



- S: flights that never entered a group.
- G: group time of the flights that entered one.
- A: the solo stretches of those same flights, same pilots, wings, loggers and days.

A separates the group from who flies in groups: same pilots, no group. A *piece* is an uninterrupted stretch of one population.

Which flights: the open ones



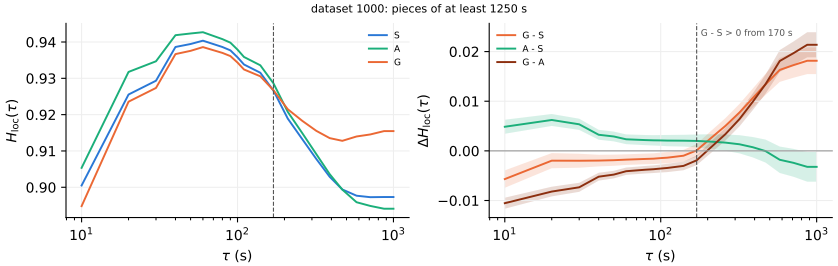
Opening. distance takeoff to landing over the path flown: 1 straight, ≈ 0 back at the takeoff. We keep opening ≥ 0.1: 55 879 of the 66 845 declared-open flights (84%).

The longer the lag, the more folded the flights (median opening 0.32, then 0.23): that is why solo MSD bends at long lags.

Question 1

Do flights in a group move differently from solo flights?

Raw: group time is straighter at the long lags



- $H_{\mathrm{loc}}(\tau)$: how straight the flight is at the time lag τ; 1 for a straight line, $\frac{1}{2}$ for a random walk.
- $H_{\mathrm{loc}}(1000\ \mathrm{s})$: G 0.915, S 0.897, A 0.894.
- $\mathrm{G} - \mathrm{S} = +0.018\ [+0.015, +0.021]$; -0.006 at 10 s, positive from 170 s.

Open flights, pieces of at least 1250 s (dataset 1000): 11 077 flights in G, 8 991 in A, 43 977 in S. 90% bands: bootstrap of pilots and blocks.

→ how we compute H_{loc} → how the bands are built → longer lags

The gain starts after about one leg of the flight

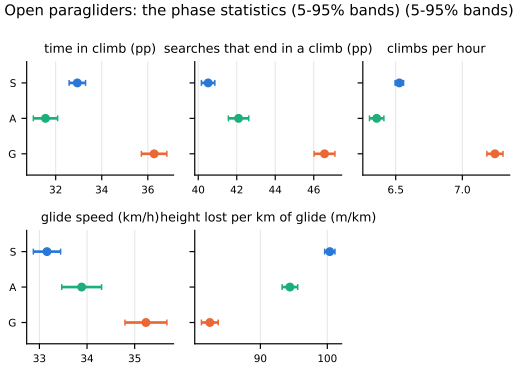
170 s	≈ 200 s	≈ 155 s
G − S turns positive	median glide, in G and in S	median climb

band above 0 from 210 s

- Within the first leg group time is a little less straight (-0.006 at 10 s); beyond it, straighter.
- The same flight gains earlier: $g > 0$ from 50 s.
- Glides last as long in G; G finds the next climb more often: 46.6% of its searches against 40.5%.

Why it pays after a first leg is open: waiting to form at the start, a better search for thermals, known thermals, or zones and weather.

Raw: group time also flies differently



G climbs more, finds the next climb more often, glides faster and loses less height; A stays close to S. Each flight counts once; the bands of G − S exclude 0 for all seven statistics.

→ how each flight counts → the seven statistics

Question 2

Is it the group, or what goes with it?

The raw gap mixes everything

Group time differs from solo time in

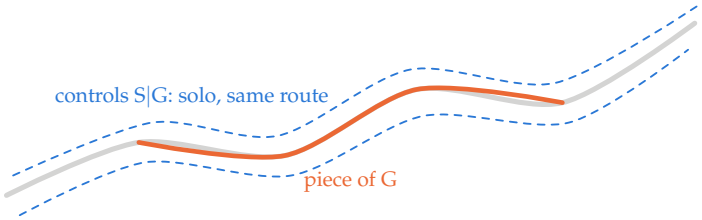
- the **route** flown,
- the **season** and the **hour**,
- the **wing**,
- the pilots who fly in groups.

We take them apart

1. fix them, and see what is left;
2. split the gap among them.

Same route, season, hour and wing: does the group still make a difference?

Same route, season, hour and wing



route: 80% of the piece within 500 m

another date, ±2 months

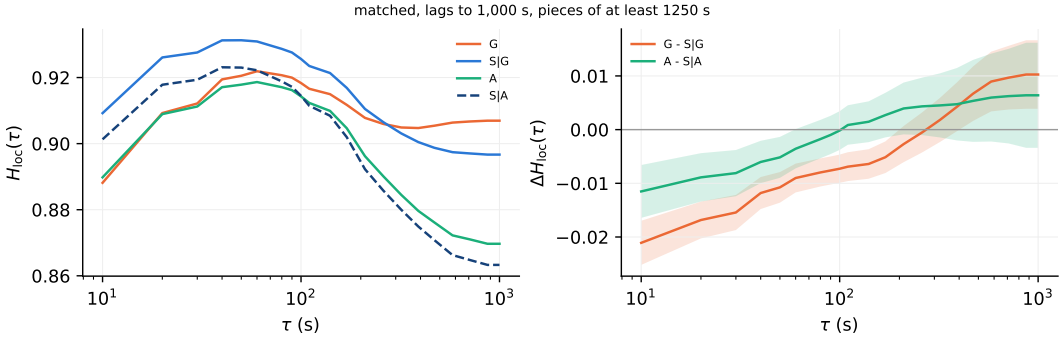
hour of the day ±2 h

same wing class

- Each piece of G is compared with solo segments on its route.
- Controls must match within tolerances, the *calipers*: the first of four sets that keeps 200 pieces per wing level is month ±2, hour ±2, same class: 2 717 pieces of G, 2 474 flights, 1 378 pilots.
- A against its own controls S | A measures who flies in groups.

→ what is a control → why the long dataset stops at 2760 s

On the same route the gap shrinks, but stays

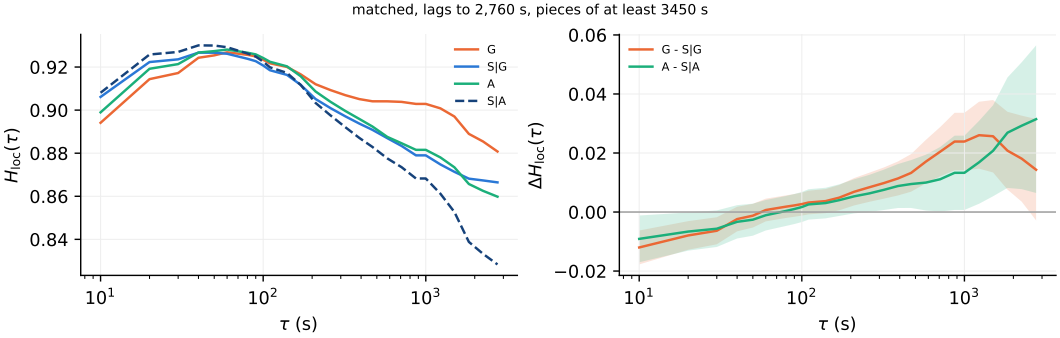


- $\mathrm{G}-\mathrm{S}|\mathrm{G} = +0.010\ [+0.004, +0.017]$ at 1000 s, against $+0.018$ raw.
- Selection $\mathrm{A}-\mathrm{S}|\mathrm{A}$: $+0.006\ [-0.003, +0.016]$.

Pieces of at least 1250 s; left: H_{loc} of G, its controls S|G, A and S|A; right: the differences with 90% bands.

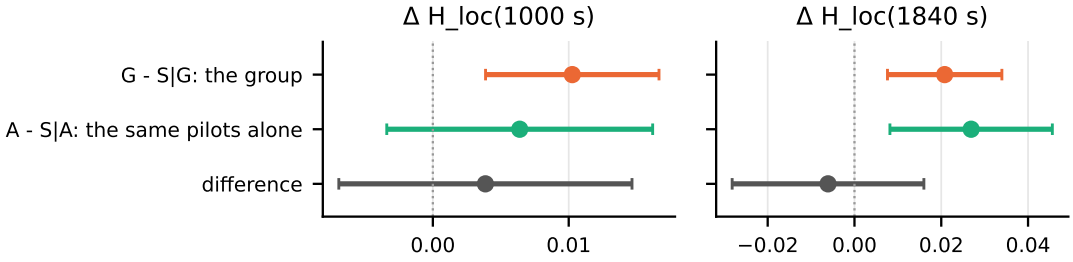
→ the nine quantities → robustness

On longer pieces it grows with the lag



- A lag τ needs pieces of at least $\tau/0.8$: for 1840 s, pieces of at least 3450 s.
- $\mathrm{G}-\mathrm{S}|\mathrm{G} = +0.021$ $[+0.008, +0.034]$ at 1840 s; at 1000 s on the same pieces $+0.024$ $[+0.014, +0.034]$.

But part of the gap is who flies in groups



$A - S|A$: the same pilots alone against their controls. At 1840 s they already show the gap: on the same route the group effect is not separable from who flies. We need the same flight: question 3.

On the same route the group climbs more and glides slower

+3.0 pp
more time in climb
[+1.4, +4.5] percentage points

+0.73
more climbs per hour

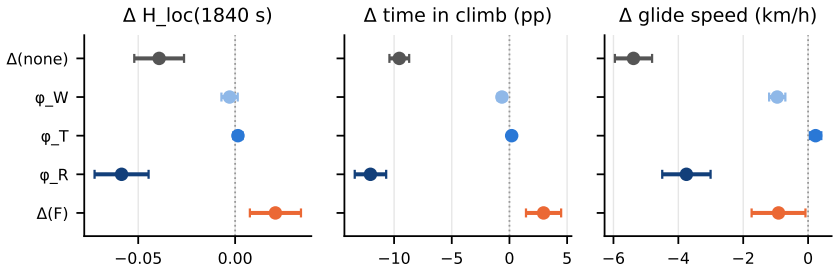
−0.9
km/h of glide speed

+6
m more height lost per km

Same route, season, hour and wing: group time climbs more, glides slower and steeper. At 1000 s the same pilots alone do not differ from their controls (5 of the 9 quantities include 0: two H_{loc} and seven phase statistics, five of them here). The raw faster glide of G (+2.1 km/h) changes sign on the same route.

How much does each factor explain?

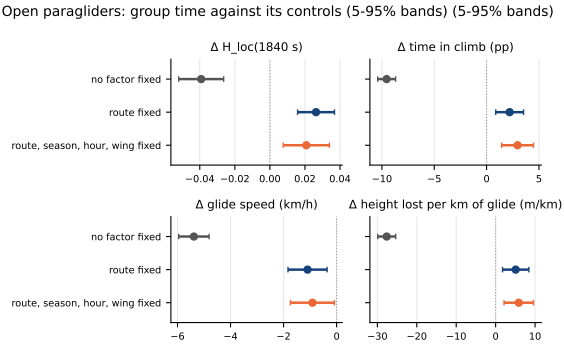
Open paragliders: the raw gap of group time and its parts (5-95% bands)



Shapley split. route R, season and hour T, wing W: $\Delta(\varnothing) = \Delta(F) + \varphi_R + \varphi_T + \varphi_W$. $\Delta(\varnothing)$ ("none") has controls from any route, $\Delta(F)$ all fixed.

→ all nine quantities → the eight cells and their curves

The route is almost all of it

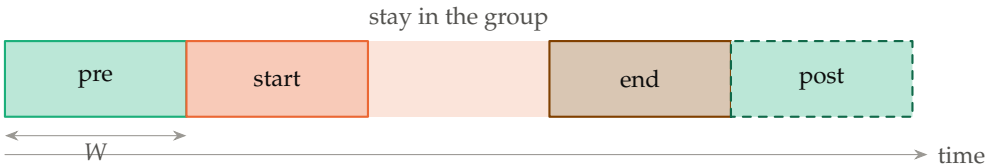


The route carries most of what the factors explain (98% at $H_{\mathrm{loc}}(1840\ \mathrm{s})$, 96% for time in climb, 84% for glide speed); season and hour almost nothing, the wing little. Fixing the route is enough.

Question 3

Does the same flight change when it enters a group, and when it leaves?

The same flight, before, during and after

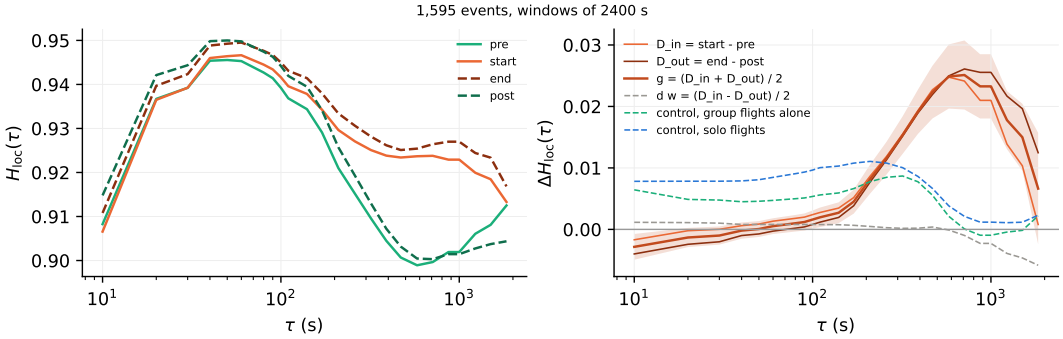


$$D_{\text{in}} = \text{start} - \text{pre}, \quad D_{\text{out}} = \text{end} - \text{post}, \quad g = \tfrac{1}{2}(D_{\text{in}} + D_{\text{out}}), \quad d\,w = \tfrac{1}{2}(D_{\text{in}} - D_{\text{out}})$$

- Same pilot, glider, logger and day: it removes who flies and when.
- g: what the group does; $d\,w$: the drift of the flight over time. An *event*: one group stay with the four windows.
- Controls: the same differences on windows with no group.

Windows of $W = 2400$ s, lags up to 1840 s; open flights.

With windows of 2400 s the flight is straighter in the group

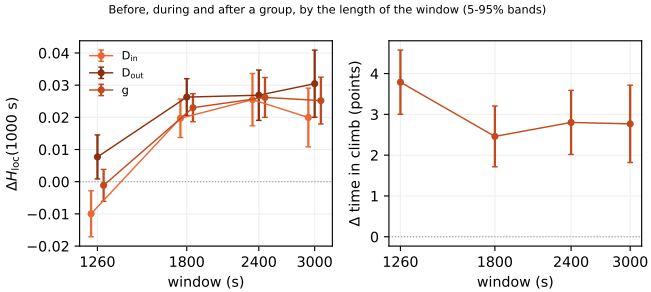


- $g = +0.026$ $[+0.020, +0.032]$ at $H_{\text{loc}}(1000\ \text{s})$; $D_{\text{in}} = +0.025$, $D_{\text{out}} = +0.027$.
- Time in climb $+2.8$ $[+2.0, +3.6]$ pp; the control pairs stay near 0.

1 595 events with the four windows.

→ all the differences

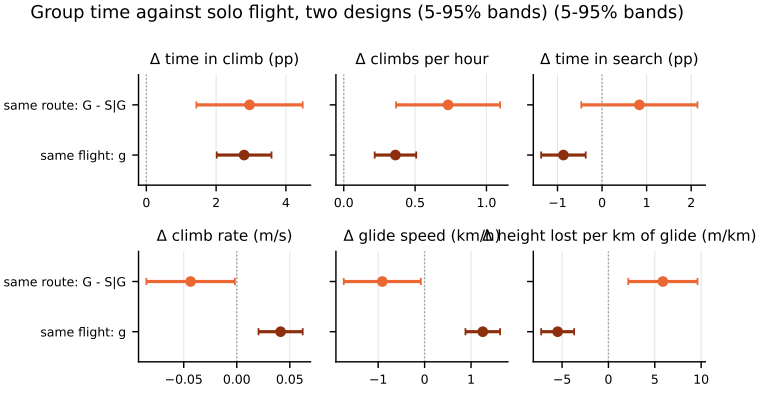
g is stable from 1800 s; at 1260 s it vanishes



- g at $H_{\text{loc}}(1000\ \text{s})$: -0.001 (1260 s), $+0.023$ (1800), $+0.026$ (2400), $+0.025$ (3000).
- At 1260 s the lag 1000 s is at the limit of the $\tau/0.8$ rule: only 260 s of the window can start a 1000-s displacement.
- Time in climb: $+2.5$ to $+3.8$ pp at every length.

Events with the four windows: 3 549, 2 567, 1 595 and 695.

Same route and same flight: some agree, some do not



Inside its group a pilot glides faster and flatter than before and after; on the same route group time glides slower and steeper. We give no reason.

Take-home

1 In a group the flight is straighter at the long lags

2 On the same route the gap halves; at long lags the pilots who fly in groups show part of it

3 The same flight, which removes who flies, is straighter inside its group

$+0.018$
$H_{\mathrm{loc}}(1000\ \mathrm{s})$, raw

$+0.010$
$H_{\mathrm{loc}}(1000\ \mathrm{s})$, same route

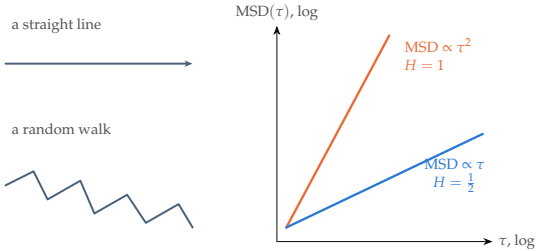
$+0.026$
$H_{\mathrm{loc}}(1000\ \mathrm{s})$, g

Open: glide speed and height lost point the other way in 3 than in 2.

Appendix

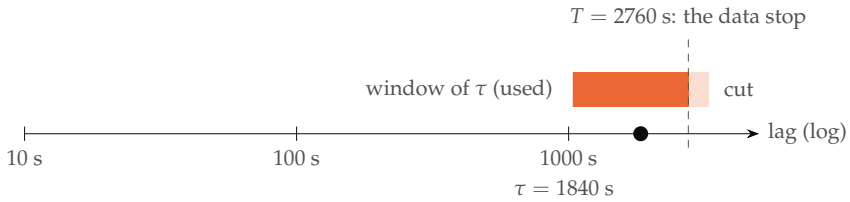
Details, and answers to questions

How we compute H_{loc} in practice



- Per flight, the time-averaged MSD on the 10-s grid of its pieces; the ensemble MSD is the mean over flights, one weight per flight.
- $\text{MSD}(\tau) \sim \tau^{2H}$ on a geometric grid of lags; $H_{\text{loc}}(\tau) = \frac{1}{2} \times$ least-squares slope of $\log \text{MSD}$ against $\log \tau$ over the lags within ± 0.25 decades of τ (from $\tau/1.78$ to $1.78\,\tau$); with fewer than 3 lags, the 3 nearest.
- H: the same fit over all lags, from 10 s to the end of the dataset.

H_{loc} at the edge of the lags: a one-sided estimate



- At the longest lags T of a dataset the window is cut on its upper side: the slope uses only lags below T, so it is centred at a shorter lag than τ and is asymmetric.
- Example: $H_{\mathrm{loc}}(1840\ \mathrm{s})$ on the dataset to $T = 2760$ s uses 1035 to 2760 s instead of 1035 to 3272 s; $H_{\mathrm{loc}}(1000\ \mathrm{s})$ on the dataset to 1000 s uses 562 to 1000 s, but 562 to 1778 s on the dataset to 2760 s: complete.
- All curves of a dataset share T, hence the same cut: differences between populations are like for like, and every bootstrap draw is cut the same way, so the bands include it.

The phase statistics: how each flight counts

Example, glide speed = distance of glides / time in glides, for two flights:

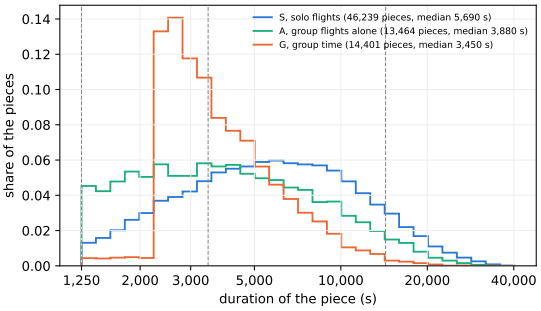
flight	glide distance	glide time	flight time	glide speed
1	20 km	0.5 h	1 h	40 km/h
2	60 km	2 h	4 h	30 km/h

Ratio of sums. $\dfrac{20+60}{0.5+2} = 32$ km/h. Flight 2 weighs four times flight 1: it glides four times longer. The longest flights dominate.

Ours: average over flights, then ratio. $\dfrac{\text{mean}(20/1,\ 60/4)}{\text{mean}(0.5/1,\ 2/4)} = \dfrac{17.5}{0.5} =$ 35 km/h. A flight weighs by its share of time gliding, not by its length.

- With a unit u: $r_u = n_u/T_u$, $q_u = d_u/T_u$, stat $= \sum_u w_u r_u / \sum_u w_u q_u$; n_u, d_u summed over its runs, T_u its flight time.
- The MSD also gives one weight per flight. On the data: time in climb of S 32.9% (ours), 34.2% (ratio of sums).

Datasets and the $\tau/0.8$ rule



A piece of ℓ s gives $(\ell - \tau)/10 + 1$ pairs of origins. Requiring $\ell \geq \tau/0.8$ gives the same share of origins at the longest lag.

- Dataset to T: pieces of at least $T/0.8$ s, at every lag up to T.
- 1000: pieces of 1250 s; 2760: 3450 s; longest (11 450 s): 14 312 s.
- Pieces of G are shorter: a group lasts at least $\tau^* = 38.1$ min.

Why the long dataset stops at 2760 s

caliper (month, hour, wing)	$T = 2250$	2760	3380
±1, ±1, class	311/395	166/231	74/111
±1, ±1, level	327/656	176/372	79/189
±2, ±2, class	513/725	**279/421**	129/228
±2, ±2, level	537/1095	294/650	139/370

pieces of G with a control, beginners/experienced; the rule asks for 200 in each level.

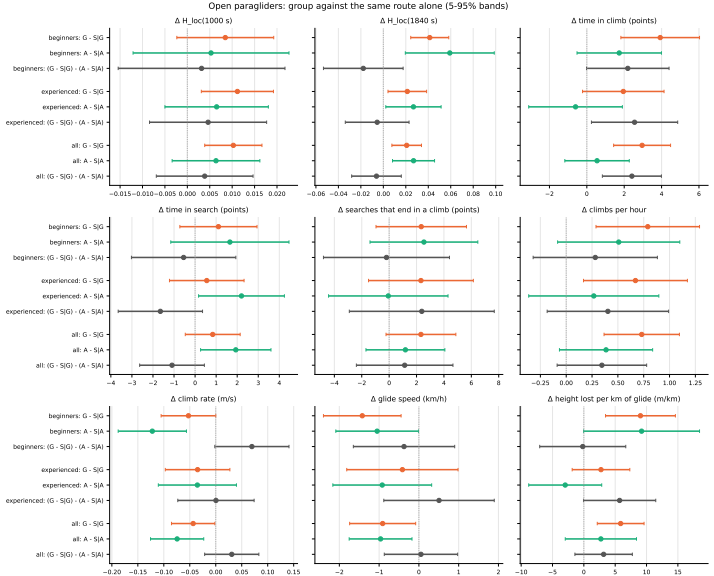
- $H_{\mathrm{loc}}(1840\ \mathrm{s})$ uses the lags within ±0.25 decades: 1035 to 3272 s.
- At $T = 2760$ the window is cut at 2760 s; at 3380 s no caliper reaches 200 pieces per level.

Matching: what is a control

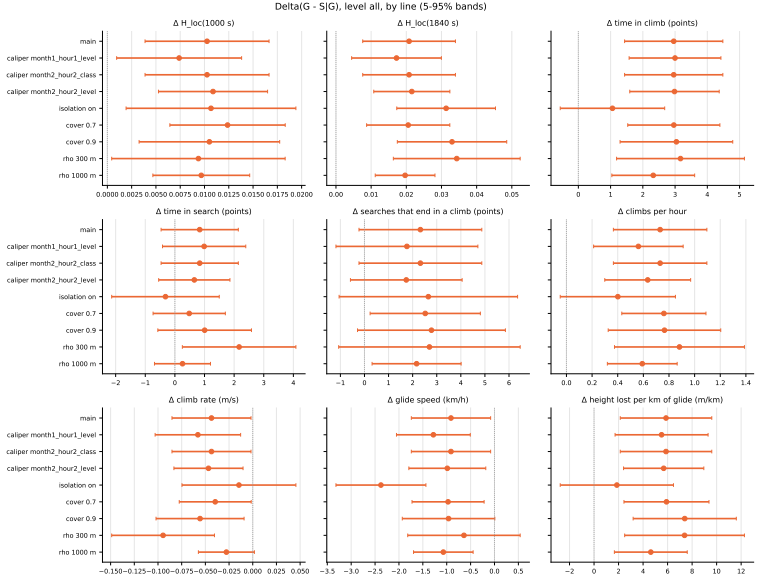
- Candidates: open flights with no contact of any length, on another date.
- Coverage: at least 80% of the points of the piece, one every 30 s, have a pool track point within 500 m, with the same direction.
- Calipers, tried in order: month ± 1 or ± 2, hour ± 1 or ± 2, wing class or level; the chosen one is the first with 200 pieces of G per level.
- Each unit is reweighted to the piece, so a piece with many controls counts once.

Isolation (nobody within 10 km) is off in the main result and is one of the robustness lines.

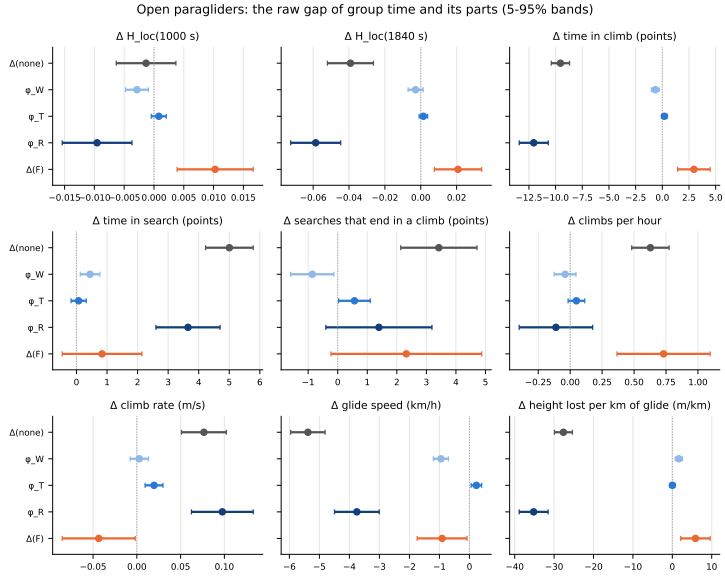
Same route: the nine quantities, by wing level



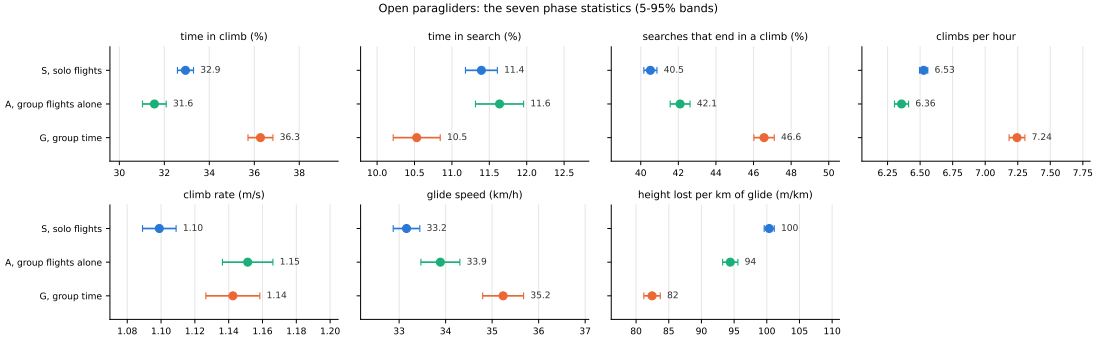
Same route, 9 lines (calipers, isolation, cover, route radius): $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ **from** $+0.007$ **to** $+0.012$**,** $\Delta H_{\mathrm{loc}}(1840\ \mathrm{s})$ **from** $+0.017$ **to** $+0.034$



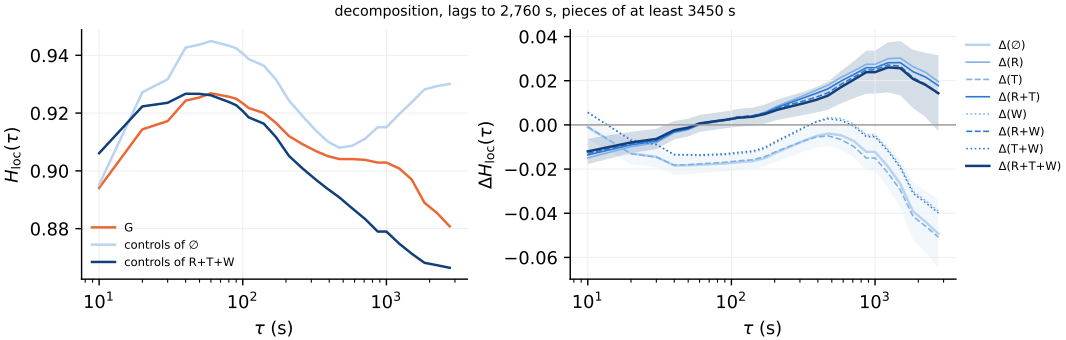
The decomposition: all nine quantities



Raw: the seven phase statistics



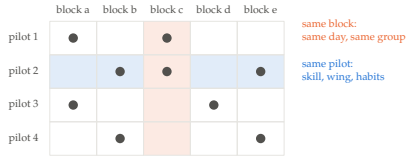
The controls of the decomposition



- Eight cells: each factor fixed or not; $\varphi_R, \varphi_T, \varphi_W$ their Shapley values.
- Controls without the route: windows of solo pieces at least as long as the piece, so from long flights; they climb more and glide faster than S.

Cell ∅: no factor; *F*: route, season and hour, wing.

Where the uncertainty comes from



- The archive is one sample: other seasons would bring other pilots and other days, and every number would change.
- Flights are not independent: those of one pilot share skill, wing and habits; those of one block share the day and the group.
- Pilots and blocks cross: a pilot flies in many blocks, a block holds many pilots.

Block. flights tied by an *episode* of the group identification (groups linked in time through a shared member), or by a *site-day* (takeoffs within 3 km on one day); chained into connected components.

A cell is one pilot in one block, about one flight.

The band: three bootstraps, one formula

- Draw the pilots with replacement; each unit weighs the count of its pilot; recompute the statistic. Its variance over the draws is V_P.
- Same for the blocks (V_B) and for the cells (V_C): 1 000 draws each.

Variance and band. units of one pilot or one block covary, so

$$V = \max\big(V_P + V_B - V_C,\ V_P,\ V_B\big), \qquad \text{band} = \hat{\theta} \pm 1.645\sqrt{V} \quad (90\%).$$

V_C is subtracted once: the pairs of one cell are in both V_P and V_B.

- All the numbers of a section use the same draws: a difference has the band of its own draws.
- Pieces and their controls are drawn together, control flights included: a control that serves many pieces moves them all at once.

Cameron, Gelbach and Miller, J. Bus. Econ. Stat. 29, 238 (2011).

Does the band hold? A check by simulation

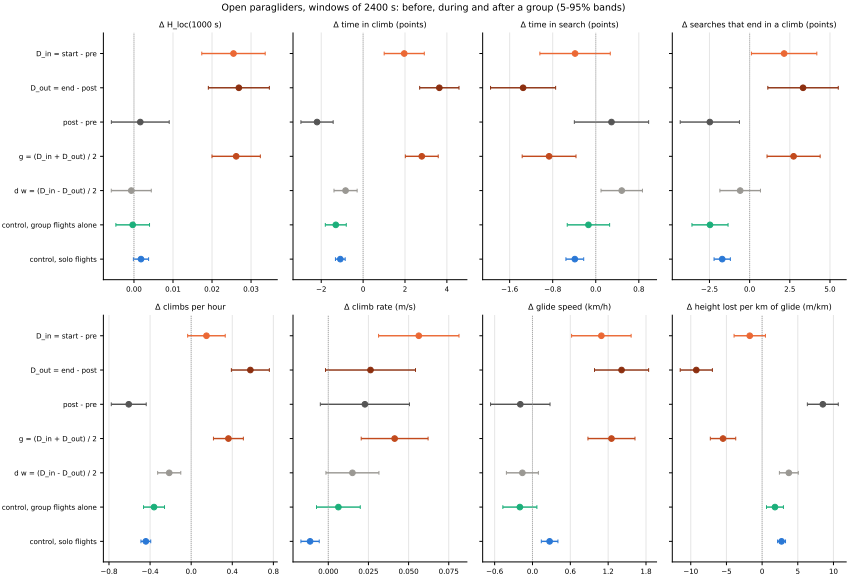
- On the real designs, replace every value by pilot + block + flight + unit effects: no group effect.
- A 90% band must hold 0 in 90% of the replicates.
- The two-way band does so in 88–99%; the bands used before did not (59% when pilots matter).

	raw		same route		same flight	
	new	old	new	old	new	old
unit only	91	90	99	100	90	88
flight only	92	91	89	100	90	90
half flight, half unit	88	85	92	100	88	88
pilot strong	88	59	88	100	89	81
block strong	88	84	88	99	89	88

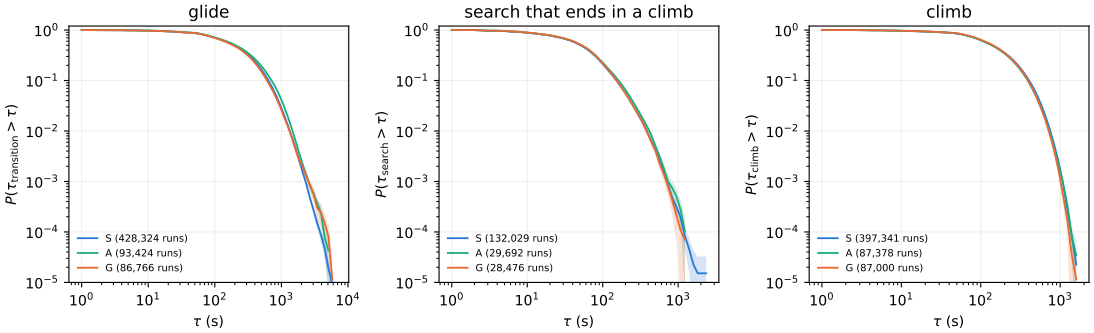
% of replicates whose 90% band holds 0.

Left out of the band: the rules of the matching (varied in the robustness lines) and the dependence on one year (planned).

Before and after, windows of 2400 s: all the differences

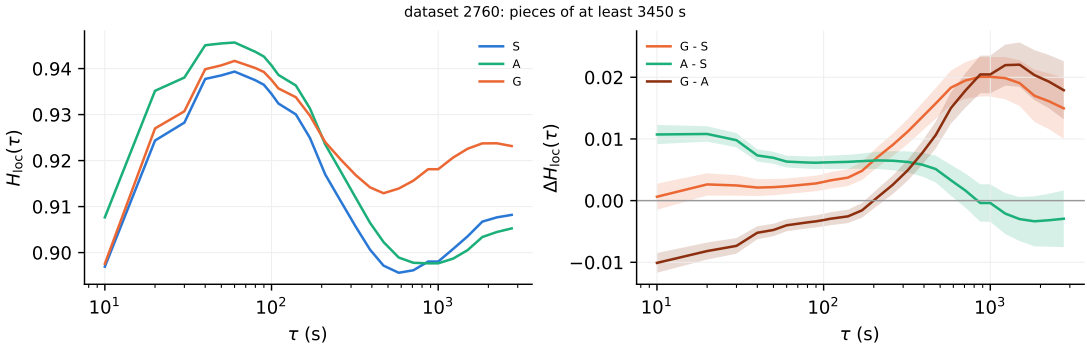


Raw: the durations of the phases



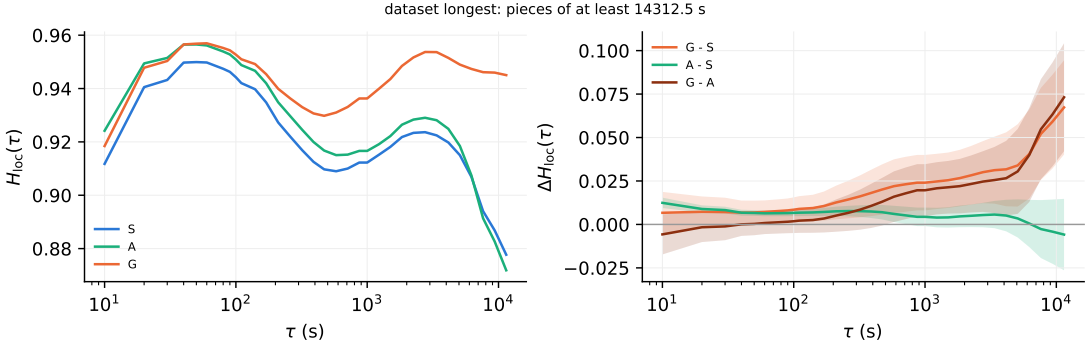
- Survival $P(\tau_{\text{phase}} > \tau)$ of climbs, searches that end in a climb and glides.
- The three populations overlap for climbs and searches; the tail of S falls below for glides.

Raw at longer lags: dataset to 2760 s



Pieces of at least 3450 s.

Raw at the longest lags: dataset to 11 450 s



- H_{loc} of S and A falls at the longest lags, that of G does not.
- G keeps few flights there: pieces of at least 14 312 s.

Variability by year

- The bands are the only uncertainty drawn.
- The estimates with one year left out are stored for every result.
- Planned: a study of how the effects change from one year to another.